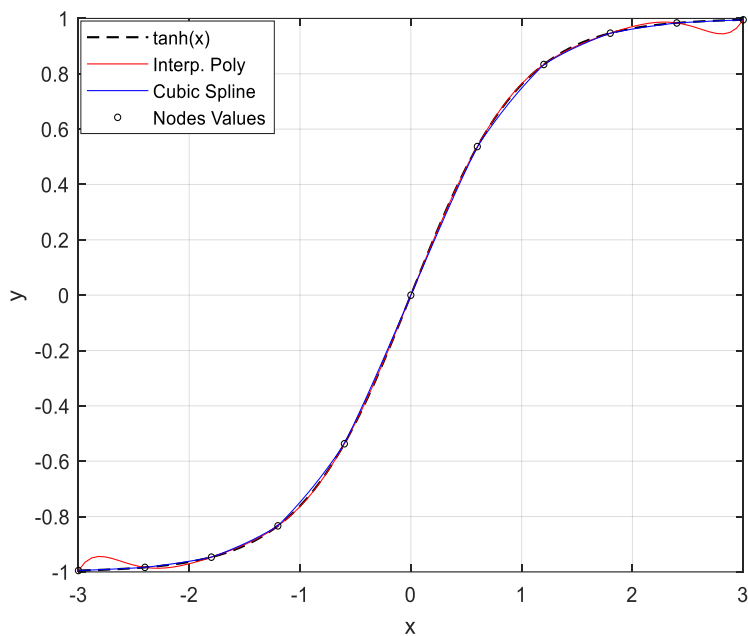
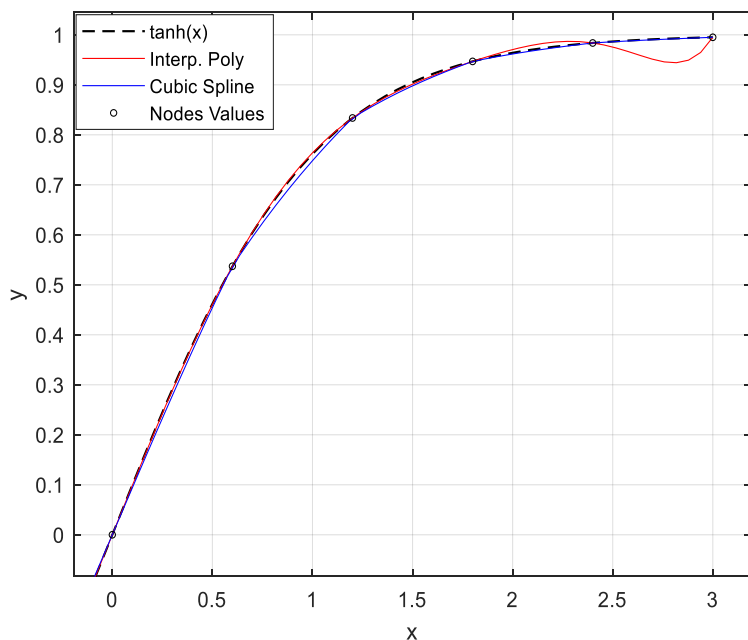


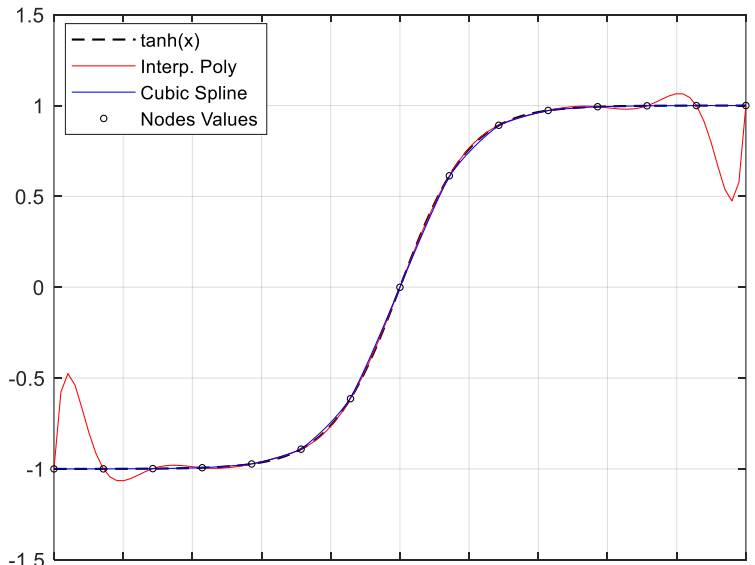
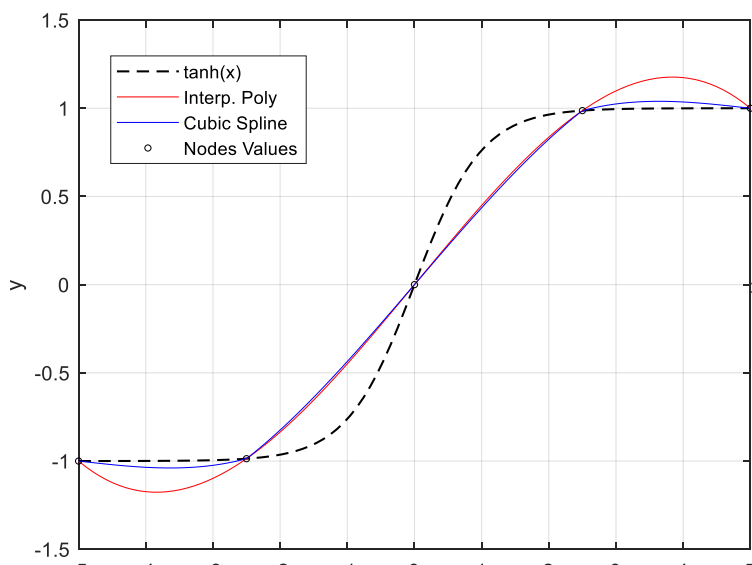
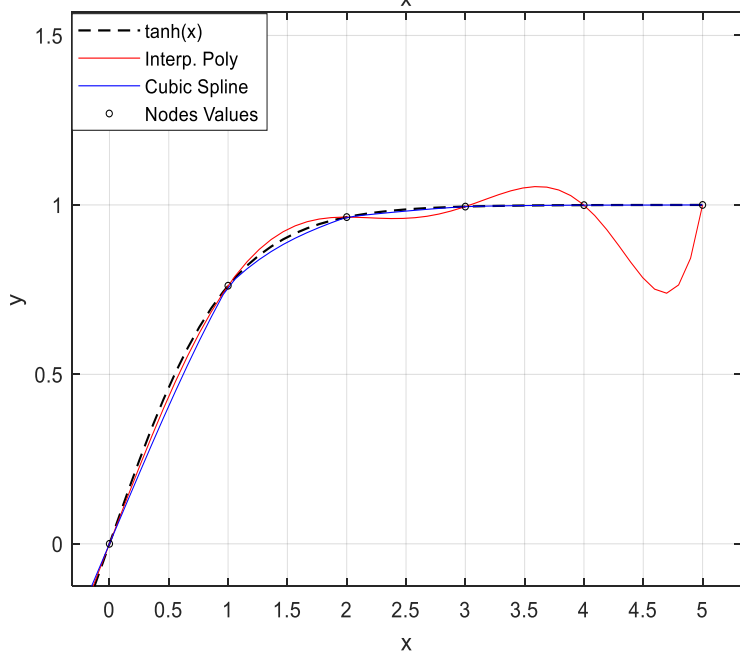
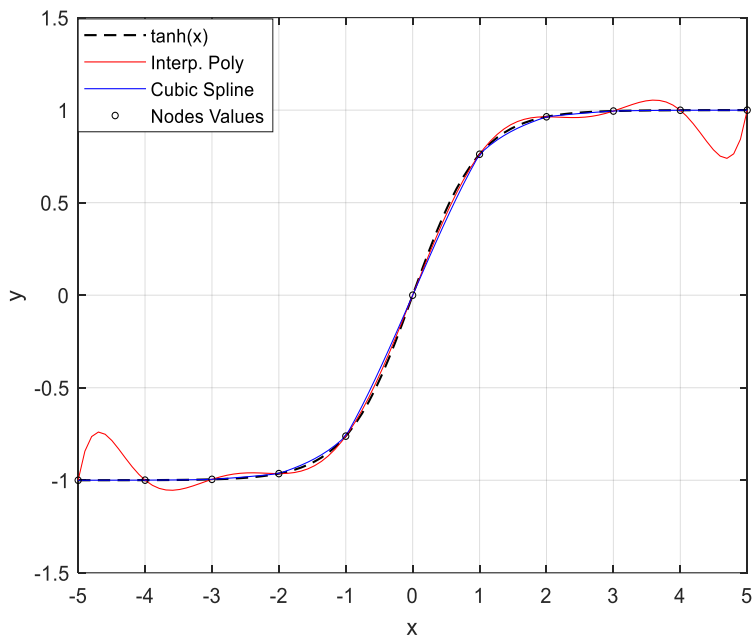
Project #6 Report by Tomas Salaz
November 15, 2024

Task: Approximate the Function $u_0(x) = \tanh(x)$ on the interval $[-3,3]$ by using 11 equally spaced nodes and constructing $P_n(x)$. Then repeat the process using $S(x)$, Cubic Splines. Plot the Original Function, Cubic Spline, and Interpolating Polynomial and Describe Findings.



Pictured to the left are the Functions asked for and their nodes (see Legend) on the interval $[-3,3]$ (top) and $[0,3]$ (bottom) for a better look at the end behaviors of the Functions. While both approximations are accurate, the Natural Cubic Spline and the Original Function are nearly indistinguishable while the Classic Interpolating polynomial is pretty accurate outside of the last node.





Pictured to the left are the same Functions with the same number of nodes, with the only change being the interval. Interval $[-5, 5]$ (top) and Interval $[0, 5]$ (middle) show The Cubic Spline is accurate no matter the interval and that the classic Interpolating Polynomial is accurate, this time, outside of the last 2 Nodes.

To that this, I tried increasing and decreasing the nodes. The Bottom Left has 5 nodes, and the Bottom Right has 15 nodes, both with equal spacing. The 5 Node figure shows that the Cubic Spline and Interpolating Polynomial are nearly identical on the interior points while the increasing the nodes decreased accuracy for Interpolation while it increased for the Spline. The Classic interpolation also keeps the Oscillation near the end points.

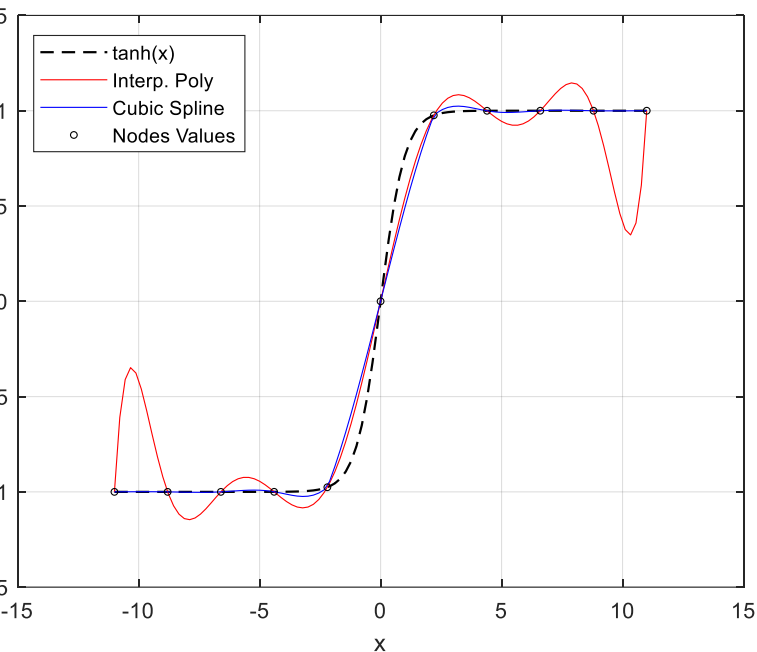
Testing the Theories that

1.Cubic Splines and Classic Interpolations are nearly identical over a large interval with lesser nodes, except for the endpoints

2.Classic Interpolations keep oscillations near ends regardless of Nodes and interval

3.A higher number of Node increased spline accuracy

Is done in the Figures below.



The Top Figure uses 11 nodes on $[-11, 11]$, the Bottom Left uses 5 nodes on the same interval and the Bottom Right uses 15 nodes. The most noticeable and useful observation is that the Cubic Spline is the most accurate with an increase in nodes but as the nodes decrease it resembles Classic Interpolation. For the work that is done to attain the Cubic Spline, it seems like a harsh trade-off for smaller applications while the Cubic Spline should be used when accuracy is needed over the whole interval.

